Thermodynamic Cost of Information Processing in Quantum Circuits

Lindblad Bounds on Gates, Error Correction, and Fault-Tolerant Depth

kds1123

Working draft, 8 November 2026

Abstract

We derive thermodynamic lower bounds on the cost of running quantum circuits using only Lindblad (GKSL) dynamics, the quantum relative entropy, and the second law in entropy-production form. Four results are established. (i) A local second law: for a circuit implemented by a time-dependent Lindbladian with locally detailed-balanced dissipators, entropy production is a sum of non-negative terms, one per dissipative channel, so every fault location and every reset carries its own bound. (ii) Without error correction, the relative entropy to the maximally mixed state contracts exponentially in depth, so noisy circuits lose all information after depth $O(\log N/\gamma\tau)$. (iii) Any error-correction round must export at least $kT\,H(s)$ of heat, where $H(s)$ is the Shannon entropy of the syndrome distribution. This yields an entropic ceiling on tolerable noise ($\approx 18.9\%$ depolarizing at zero rate), a thermal-threshold condition $T \le \hbar\omega/[k\ln(1/\varepsilon_{\text{th}}-1)]$ on the reset bath, and a depth law $Q_{\min} \propto N_L D_L\, d^3 s(p)$ with $d = \Theta(\log(N_L D_L))$ for surface-code fault tolerance. (iv) Numerically, the Landauer floor for a million-logical-qubit machine is tens of nanowatts at the wall plug, many orders of magnitude below any engineering estimate. The binding constraints are entropy throughput and reset fidelity, not Landauer heat.

1 Introduction

Discussions of the thermodynamics of computation often stop at the observation that erasing a bit costs $kT\ln 2$. For quantum circuits this is too coarse in three ways. First, ideal unitary gates are logically reversible and carry no Landauer cost, so all irreversibility sits in noise, measurement, and reset. Second, error correction converts noise entropy into syndrome entropy that must be physically removed every round, so the cost scales with the number of fault locations, not the number of logical operations. Third, fault-tolerance thresholds are statements about noise strength, and at fixed level spacing noise strength is a statement about temperature; thresholds therefore translate into temperature and entropy-flux constraints.

The contributions are:

1. A local form of the second law for Lindbladian circuits (Theorem 1) and the resulting Landauer, free-energy, and finite-time bounds (Propositions 1–4).
2. An entropy–error tradeoff for bath-assisted operations (Proposition 5).
3. A depth limit for noisy circuits without error correction, expressed as relative-entropy contraction (Proposition 6).
4. A syndrome-entropy Landauer bound for error-correction rounds, an entropic ceiling on noise, and a thermal threshold condition (Theorem 2, Corollary 1, Propositions 7–8).

5. A scaling law for the minimum dissipation of fault-tolerant computation in circuit depth and width, with a numerical estimate (Theorem 3).

Entropies are in nats, k_B is written explicitly where energies appear, and $S(\rho) = -\operatorname{Tr}\rho\ln\rho$.

2 Framework

2.1 Lindblad dynamics

Consider N qubits with a controlled system Hamiltonian $H_S(t)$. The state evolves as

$$\dot\rho = -i[H_S(t),\rho] + \sum_k \mathcal{D}_{k,t}(\rho), \tag{1}$$

$$\mathcal{D}_{k,t}(\rho) = \sum_j \gamma_{kj}\Big(L_{kj}\rho L_{kj}^\dagger - \tfrac{1}{2}\{L_{kj}^\dagger L_{kj},\rho\}\Big). \tag{2}$$

We assume:

- **(A1)** Markovian GKSL form, so each $e^{s\mathcal{D}_{k,t}}$ is completely positive and trace preserving.
- **(A2)** Local detailed balance: each $\mathcal{D}_{k,t}$ has a stationary state $\sigma_k(t)$. For a thermal bath at inverse temperature β_k, $\sigma_k(t) = e^{-\beta_k H_S(t)}/Z_k(t)$; for unital noise (depolarizing, dephasing) $\sigma_k = I/2^N$.
- **(A3)** Gate times are long compared with the bath correlation time, which justifies using the instantaneous Hamiltonian in the dissipator. Outside this regime the master equation is not internally consistent (Alicki–Lidar–Zanardi [25]), and the bounds should be read in the system-plus-bath unitary picture of Section 3.

The heat current into the system from bath k is $\dot Q_k = \operatorname{Tr}[H_S\,\mathcal{D}_k(\rho)]$.

2.2 Relative entropy and entropy production

The quantum relative entropy $S(\rho\|\sigma) = \operatorname{Tr}\rho(\ln\rho - \ln\sigma)$ is non-negative, unitarily invariant, jointly convex, and monotone under CPTP maps. Define the entropy production rate

$$\dot\Sigma = \frac{dS}{dt} - \sum_k \beta_k \dot Q_k. \tag{3}$$

Theorem 1 (Local second law)**.** *Under (A1)–(A2),*

$$\dot\Sigma = \sum_k \dot\sigma_k, \qquad \dot\sigma_k = -\operatorname{Tr}\big[\mathcal{D}_k(\rho)\big(\ln\rho - \ln\sigma_k\big)\big] \;\geq 0. \tag{4}$$

Proof. The Hamiltonian part does not change the von Neumann entropy, so $dS/dt = -\operatorname{Tr}[\sum_k \mathcal{D}_k(\rho)\ln\rho]$. For thermal σ_k, $\ln\sigma_k = -\beta_k H_S - \ln Z_k$ and $\operatorname{Tr}\mathcal{D}_k(\rho) = 0$, so $-\beta_k\dot Q_k = \operatorname{Tr}[\mathcal{D}_k(\rho)\ln\sigma_k]$. Summing gives the stated form. For non-negativity, $e^{s\mathcal{D}_k}$ is CPTP with fixed point σ_k, so $s\mapsto S(e^{s\mathcal{D}_k}\rho\|\sigma_k)$ is non-increasing for $s\geq 0$. Its derivative at $s=0$ is $\operatorname{Tr}[\mathcal{D}_k(\rho)(\ln\rho-\ln\sigma_k)]$, hence $\dot\sigma_k\geq 0$. □

The consequence used throughout is additivity: total entropy production of a circuit is a sum over gate locations, idle periods, measurements, and resets, each separately non-negative. Any subset of terms is therefore a valid lower bound on the total.

3 Generalized Landauer Bounds

Proposition 1 (Lindblad–Landauer)**.** *Let $Q_k^b = -\int \dot{Q}_k\, dt$ be the heat delivered to bath k over a protocol taking ρ to ρ'. Then*

$$\sum_k \beta_k Q_k^b = \Sigma - \Delta S \;\geq\; S(\rho) - S(\rho'), \tag{5}$$

with equality iff every $\dot{\sigma}_k$ vanishes along the trajectory (quasi-static operation). The bound applies to any logically irreversible map, including discarding ancillas, not only bit erasure.

Proposition 2 (Exact form beyond Lindblad)**.** *Let system A be coupled by a global unitary to a bath B initially thermal at β and uncorrelated with A. Then (Reeb–Wolf [12], rearranged)*

$$\beta Q = S(\rho_A) - S(\rho'_A) + I(A' : B') + S(\tau_{B'} \| \tau_B), \tag{6}$$

where Q is the heat absorbed by B, I is the post-interaction mutual information, and τ_B is the initial thermal state. Every term beyond the first is non-negative, so $\beta Q \geq S(\rho_A) - S(\rho'_A)$ with no weak-coupling or Markov assumption.

Proposition 3 (Free-energy ledger)**.** *Define $F_{\mathrm{neq}}(\rho, H) = \mathrm{Tr}\,\rho H - kT\, S(\rho) = F_{\mathrm{eq}}(H) + kT\, S(\rho \| \gamma_H)$, with γ_H the Gibbs state. For a protocol with work W and $H_0 \to H_\tau$, integrating Theorem 1 gives*

$$\beta W = \beta \Delta F_{\mathrm{eq}} + \big[S(\rho_\tau \| \gamma_\tau) - S(\rho_0 \| \gamma_0) \big] + \Sigma. \tag{7}$$

For a thermal initial state the dissipated work obeys $\beta W_{\mathrm{diss}} \geq S(\rho_\tau \| \gamma_\tau)$.

Proposition 4 (Finite-time Landauer)**.** *In the slow-driving regime $\Sigma = \int \dot{\lambda}^\top \zeta(\lambda) \dot{\lambda}\, dt$, with ζ a positive semi-definite friction tensor proportional to the bath relaxation time. By Cauchy–Schwarz,*

$$\Sigma \;\geq\; \frac{\mathcal{L}^2}{\tau}, \qquad \mathcal{L} = \int \sqrt{\dot{\lambda}^\top \zeta\, \dot{\lambda}}\; dt, \tag{8}$$

where $\mathcal{L}$ is the thermodynamic length [11, 14]. Combined with Proposition 1,

$$Q^b \;\geq\; kT \big[S(\rho) - S(\rho') \big] + kT\, \frac{\mathcal{L}^2}{\tau}. \tag{9}$$

Remark (Single-shot costs). The von Neumann bounds are the typical, asymptotic limit. For a single error-correction round on a small block, costs are governed by smoothed max-relative-entropy quantities [17, 13], which for the ancilla states used below exceed the von Neumann values by terms of the order of the standard deviation of $-\ln p(s)$.

4 Gate Operations

4.1 Coherent gates

An ideal unitary U has $\Delta S = 0$, hence zero Landauer cost (Bennett [2]). The unavoidable costs are elsewhere. The control field that implements U carries finite energy, and conservation-law arguments (Ozawa [31]; Gea-Banacloche [32]) bound the gate error from below by a term scaling as $1/\bar{n}$ in the mean excitation number of the control, so control energy per gate scales as $E_c \gtrsim \hbar\omega/\varepsilon$ for target error ε, with a model-dependent prefactor. Generation and distribution of the control signal is classical and lies outside the Lindbladian.

4.2 Noisy gates

Let a gate be followed by a Pauli channel with error distribution P. Since the output is a mixture of the states $E\rho E^\dagger$,

$$S(\mathcal{N}(\rho)) - S(\rho) \le H(P), \tag{10}$$

with equality iff the error-displaced states are mutually orthogonal. For per-qubit depolarizing error of total probability p, $H(P) = s(p) := h(p) + p\ln 3$ per fault location, with h the binary entropy in nats. For small p,

$$s(p) \approx p\ln(1/p) + p(1 + \ln 3). \tag{11}$$

4.3 Entropy–error tradeoff for bath-assisted operations

Operations that use a bath actively (reset, thermalization-based gates, autonomous stabilization) are subject to Proposition 4. Their error has a decoherence floor $\varepsilon \ge \gamma_{\rm err}\tau$.

Proposition 5. *For slow-driven, bath-assisted operations with $\varepsilon \ge \gamma_{\rm err}\tau$,*

$$\Sigma \cdot \varepsilon \;\ge\; \gamma_{\rm err}\,\mathcal{L}^2. \tag{12}$$

Proof. $\Sigma \ge \mathcal{L}^2/\tau$ and $\tau \le \varepsilon/\gamma_{\rm err}$ give $\Sigma \ge \mathcal{L}^2\gamma_{\rm err}/\varepsilon$. □

Slower operation lowers dissipation and raises error, and the product has a geometric lower bound. If the same bath sets both the friction and the error rate, the product is bounded by an $O(1)$ constant in units of the squared thermodynamic distance; this is a scaling heuristic, since the two timescales are generally set by different environments. Coherent gates of superconducting or ion-trap qubits are fast compared with relaxation and are not described by Proposition 4.

4.4 Reset at finite temperature

Resetting an ancilla with level spacing $\hbar\omega$ by contact with a bath at T_c cannot produce a pure state. Full thermalization leaves a residual excited population

$$\varepsilon_r = \frac{1}{1 + e^{\hbar\omega/kT_c}}. \tag{13}$$

The Landauer price of going from the maximally mixed state to this state is $kT_c[\ln 2 - h(\varepsilon_r)] \le kT_c\ln 2$. Achieving $\varepsilon_r \le \varepsilon$ requires $T_c \le \hbar\omega/(k\ln(1/\varepsilon - 1))$, and reaching $\varepsilon \to 0$ with finite resources is forbidden by the third law [29]. Reset infidelity therefore enters the error budget as a fault location of its own, and its price is paid in refrigeration (Section 7).

5 Depth Limits Without Error Correction

Take an N-qubit circuit of D layers, each a unitary U_ℓ followed by single-qubit depolarizing noise for time τ, generated by

$$\mathcal{L}_{\rm dep}(\rho) = \gamma \sum_i \big(\,\mathrm{Tr}_i(\rho)\otimes I_i/2 - \rho\big). \tag{14}$$

The fixed point $I/2^N$ is unitarily invariant, so by Theorem 1 the quantity $D_\ell := S(\rho_\ell\|I/2^N) = N\ln 2 - S(\rho_\ell)$ is unchanged by the unitaries and non-increasing under noise.

Proposition 6. *Assume the depolarizing semigroup satisfies a modified log-Sobolev inequality with a constant $\alpha > 0$ independent of N, in the normalization $-\operatorname{Tr}[\mathcal{L}_{\mathrm{dep}}(\rho)(\ln\rho - \ln(I/2^N))] \geq 2\alpha\gamma\, S(\rho\|I/2^N)$. Then*

$$D_\ell \leq e^{-2\alpha\gamma\tau} D_{\ell-1}, \qquad D_D \leq N \ln 2 \cdot e^{-2\alpha\gamma\tau D}, \tag{15}$$

and the entropy produced in layer ℓ is exactly $\Delta S_\ell = D_{\ell-1} - D_\ell \leq (1 - e^{-2\alpha\gamma\tau}) D_{\ell-1}$. By Pinsker's inequality, the output is within trace distance η of the maximally mixed state once

$$D \;\geq\; D^* = \frac{\ln\bigl(N \ln 2/(2\eta^2)\bigr)}{2\alpha\gamma\tau}. \tag{16}$$

The existence of an N-independent constant is the content of the log-Sobolev results of Kastoryano–Temme [26] and Müller-Hermes–Stilck França–Wolf [27]; the application to noisy circuits is due to Stilck França and García-Patrón [28], and the qualitative $O(\log N)$ depth limit for unital noise without fresh ancillas is the Aharonov–Ben-Or–Impagliazzo–Nisan theorem [19]. The contraction is stated in the form needed here; no claim of optimal constant is made.

The thermodynamic reading is direct: entropy production per layer decays like $e^{-2\alpha\gamma\tau\ell}$, because noise can only destroy information still present. Beyond depth D^* the circuit is thermodynamically quiet only because the computation is gone. Non-unital noise (amplitude damping toward a Gibbs state) has a different fixed point and can lower entropy, the loophole that error correction with cold ancillas exploits [24].

6 Thermodynamics of Quantum Error Correction

6.1 Setup

Take an $[[n, k, d]]$ stabilizer code and a round consisting of: Pauli noise with error distribution P on the data block; ideal syndrome extraction by a unitary V coupling data to $m = n - k$ ancillas initialized in $|0\rangle$; recovery controlled on the ancillas; and reset of the ancillas to $|0\rangle$ by contact with a thermal bath at T_c. Let $s(E)$ be the syndrome of error E and $p(s)$ its distribution.

Theorem 2 (Syndrome-entropy Landauer bound)**.** *The heat delivered to the reset bath per round satisfies*

$$Q_c \;\geq\; kT_c\, H(s), \qquad H(s) = -\sum_s p(s) \ln p(s) = H(P) - H(E \,|\, s). \tag{17}$$

Proof. After V, the global state is $\sum_s |s\rangle\langle s|_A \otimes \bigl(\sum_{E:s(E)=s} P(E)\, E\rho_L E^\dagger\bigr)$, because the Pauli channel is a classical mixture of unitaries and distinct syndromes occupy orthogonal ancilla states. The ancilla marginal is $\rho_A = \sum_s p(s)|s\rangle\langle s|$ with $S(\rho_A) = H(s)$. The bath is initially thermal and uncorrelated with A, and the reset ends in $|0\rangle$ with $S = 0$, so Proposition 2 gives $\beta Q_c \geq H(s)$. The identity $H(P) = H(s) + H(E|s)$ holds because s is a function of E. □

Two bounds on $H(s)$ follow: $H(s) \leq H(P)$ and $H(s) \leq m \ln 2$. For a non-degenerate code that corrects all typical errors, $H(E|s) \approx 0$ and $H(s) \approx H(P)$: the entropy injected by noise is exactly the entropy exported. For degenerate codes $H(s) < H(P)$. The classical record of the syndrome, if stored and later erased, costs the same $kT\, H(s)$ and must not be counted twice. The Lindblad accounting splits the round's entropy production as $\Sigma_{\mathrm{reset}} = \beta Q_c - H(s) \geq 0$, so the bound is saturated by a quasi-static reset and exceeded by the finite-time premium of Proposition 4.

Corollary 1 (Entropy flux)**.** *In a steady state of repeated rounds at rate R with one reset per ancilla per round, $\dot{Q}_c \geq kT_c\, R\, H(s)$.*

Sustained error correction therefore requires a cold bath able to absorb an entropy flux $R\,H(s)$, the thermodynamic content of the requirement for a continuous supply of fresh low-entropy ancillas.

6.2 Entropic ceiling on tolerable noise

Syndrome entropy cannot exceed the capacity of the ancilla register, $m \ln 2$. A non-degenerate code returning all typical errors to the code space needs $H(E|s) \approx 0$, hence $n\,s(p) \lesssim (n-k)\ln 2$.

Proposition 7. *In the zero-rate limit $k/n \to 0$, a non-degenerate code under independent depolarizing noise requires $s(p) \le \ln 2$, i.e. $h_2(p) + p\log_2 3 \le 1$, which gives*

$$p \;\lesssim\; 0.1893. \tag{18}$$

This is the hashing bound for the depolarizing channel. It is a typical-set counting argument, not a strict ceiling for degenerate codes (DiVincenzo–Shor–Smolin [30]). The relevant gap is to circuit-level surface-code thresholds of order 0.5–1% [23]; between the two lies the cost of fault propagation through the syndrome-extraction circuit and of imperfect decoding.

6.3 Thermal threshold

If the dominant physical error is thermal excitation of a qubit with gap $\hbar\omega$, Section 4.4 gives $\varepsilon_r = 1/(1+e^{\hbar\omega/kT})$. Requiring $\varepsilon_r \le \varepsilon_{\rm th}$:

Proposition 8.

$$T \;\le\; \frac{\hbar\omega}{k\,\ln(1/\varepsilon_{\rm th}-1)}. \tag{19}$$

For $\varepsilon_{\rm th} = 1\%$ this is $T \lesssim \hbar\omega/(4.6\,k)$. For a 5 GHz qubit ($\hbar\omega/k \approx 0.24$ K) the condition is $T \lesssim 52$ mK. This sets the temperature of the reset bath and fixes the cold side of every refrigeration estimate below.

7 Fault-Tolerant Depth and the Landauer Floor

Let the algorithm have N_L logical qubits, logical depth D_L, and total failure budget δ. For a surface code with physical error rate p and threshold $p_{\rm th}$, the logical error per logical qubit per layer is $p_L \approx A(p/p_{\rm th})^{\lceil (d+1)/2\rceil}$, and requiring $N_L D_L p_L \le \delta$ gives

$$d \approx \frac{2\ln(AN_LD_L/\delta)}{\ln(p_{\rm th}/p)} - 1. \tag{20}$$

Each logical qubit uses $n \approx 2d^2$ physical qubits with c fault locations per qubit per round, and each logical layer takes about d rounds. The entropy exported per logical qubit per layer is therefore $\approx 2c\,d^3 s(p)$ at leading order.

Theorem 3 (Landauer floor under surface-code fault tolerance)**.** *Using Theorem 2 with $H(s) \approx H(P)$ at leading order in small p,*

$$Q_{\rm min} \approx 2c\,kT_c\,N_LD_L\,d^3\,s(p), \qquad d = \Theta\!\left(\frac{\log(N_LD_L)}{\log(p_{\rm th}/p)}\right). \tag{21}$$

Consequences: (a) the cost is linear in the space-time volume $N_L D_L$ times a factor polylogarithmic in it, $Q_{\min} \propto N_L D_L \,[\log(N_L D_L)]^3 [\log(p_{\rm th}/p)]^{-3} s(p)$; (b) it diverges as $p \to p_{\rm th}^-$ like $[\ln(p_{\rm th}/p)]^{-3}$; (c) at fixed volume it vanishes only as $s(p) \sim p \ln(1/p)$ as $p \to 0$, so entropy production per fault location is bounded below by $s(p)$ and above by $\ln 2$ per qubit per round (Proposition 7).

With ideal refrigeration from T_c to a warm environment T_h, delivering heat Q_c at T_c costs work $Q_c(T_h/T_c - 1)$, so the wall-plug floor per nat of syndrome entropy is

$$W \;\geq\; k(T_h - T_c)\, H(s) \;\approx\; kT_h\, H(s), \tag{22}$$

independent of T_c as long as $T_c \ll T_h$. Cooling the reset bath harder does not reduce the floor; it moves it from the cold plate to the compressor.

Worked estimate (order of magnitude). Take $d = 17$ (about 577 physical qubits per logical qubit), $p = 10^{-3}$, and about 10^3 fault locations per round, so $s(p) = 9.0 \times 10^{-3}$ nats and $H(s) \approx 9$ nats per round per logical qubit. With round time $1\,\mu$s, $T_c = 20$ mK, and $T_h = 300$ K: $kT_c = 2.76 \times 10^{-25}$ J, so the cold-side floor is 2.5×10^{-24} J per round, or 2.5×10^{-18} W per logical qubit. For $N_L = 10^6$ the cold-side floor is 2.5 pW; with the Carnot factor $(T_h/T_c - 1) \approx 1.5 \times 10^4$ the wall-plug floor is ≈ 37 nW.

This is many orders of magnitude below the kilowatt-to-megawatt scale of proposed fault-tolerant machines. The fundamental floor does not limit scaling; the binding constraints are cold-stage entropy throughput, reset fidelity (Proposition 8), control-line and cryogenic wiring dissipation, and the classical decoder, in line with recent energetic analyses of full-stack quantum computers [33, 34].

8 Discussion and Limits

What is bounded. The results are lower bounds on dissipation, not predictions. Theorems 2 and 3 count only syndrome entropy; they omit control energy, wiring, measurement amplification, and classical decoding, which dominate in practice.

Regime of validity. Theorem 1 and Propositions 4–6 require the Markovian master equation with instantaneous Hamiltonian, consistent only for slow gates compared with bath memory [25]. Fast coherent gates are covered by Proposition 2 and the control-energy arguments of Section 4.1, not by the friction-tensor bound.

Noise model. The syndrome-entropy results assume Pauli noise so that ρ_A is diagonal in the syndrome basis. Coherent errors, leakage, and correlated noise need separate treatment.

Asymptotic versus single-shot. The von Neumann bounds are typical-case; single-round costs involve smoothed max-entropies. A fluctuation-aware version of Theorem 3 would replace $H(s)$ by a smooth max-entropy and add a standard-deviation correction.

Open problems. (1) Tight, N-independent log-Sobolev constants for general local noise. (2) A circuit-level entropic bound explaining the factor of 20–40 between the hashing ceiling and practical thresholds. (3) A Landauer-type bound for decoding that accounts for the classical decoder alongside the syndrome record. (4) Extension of Proposition 5 to the fast-gate regime using exact bath dynamics.

9 Conclusion

The Lindblad framework with the quantum relative entropy gives a clean accounting of where quantum circuits pay thermodynamically. Ideal unitary gates are free; all Landauer-type cost sits in

noise, measurement, and reset, and Theorem 1 makes each of these a separately non-negative, locally boundable term. Without error correction, noise destroys the information in a circuit exponentially fast in depth, so entropy production ends the computation at depth $O(\log N)$. With error correction, each round must export at least $kT\,H(s)$ of heat, which scales with the number of fault locations and, under surface-code fault tolerance, as $N_L D_L\, d^3 s(p)$ with d logarithmic in the space-time volume. Fault-tolerance thresholds translate into an entropic ceiling on noise and a thermal condition on the reset bath. Numerically the resulting floor is nanowatts at the wall plug, so the practical limits are entropy throughput, reset fidelity, control dissipation, and decoding, not Landauer heat. Sharper circuit-level entropic bounds, a thermodynamics of the classical decoder, and single-shot versions of the scaling law are the main open directions.

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